

QUT Droid Racing Challenge 2025

Rules Rev 3.0 – 02/05/2025



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1. Contacts

General Inquiries

For general inquiries regarding the droid racing challenge, please contact the DRC team through the QUT Robotics club discord.

Discord

<https://discord.gg/cQa8VzVgT4>

Gmail

qutroboticsclub@gmail.com

Facebook

<https://www.facebook.com/QUTRoboticsClub/>

Instagram

<https://www.instagram.com/qut.roboticsclub>

For more information and official contact, please check out our website:
<https://qutrobotics.com/>

2. Competition Categories

- 2.1 The competition shall be split into two categories:
 - 2.1.1 Newcomers
 - 2.1.2 Opens
- 2.2 Each category shall have its own: heats, finals, scoring leaderboard, 1st Place Winner, 2nd Place Runner Up and 3rd Place Runner Up.
- 2.3 The prize money for each category may be different.
- 2.4 Each category shall have its own eligibility requirements as detailed in *Sections 3 and 4*.
- 2.5 Each category shall have slightly adjusted rules as detailed in *Section 7*. In summary:
 - 2.5.1 Newcomers races shall have the following adjustments:
 - (1) No obstacles on the track.
 - (2) No penalty for not attempting the turning challenge - i.e. if a droid fails to choose a direction to turn no penalty is applied. See Section 8 for more details.
 - 2.5.2 Opens will have the full competition rules applied including obstacles on the track and turning challenge penalties.

3. Team Eligibility

3.1 Eligible competitors for either the Newcomers or Opens category are as follows:

3.1.1 Undergraduate University Students.

3.1.2 High school students, provided they have an accompanying teacher.

- (1) This challenge is designed for undergraduate students; changes will not be made to accommodate high school students.
- (2) Highschool teams must have a dedicated staff member present for the entirety of the event and exercise control over the students.
- (3) If the high school team consists of students from multiple schools, a teacher from each school must be present.

3.2 Eligible competitors for the Opens category additionally include:

3.2.1 Higher degree students (i.e. Masters and PhD students, and Researchers and Education facility staff)

3.2.2 University students of any degree who have graduated.

3.2.3 Interested members of the public.

3.3 Teams may have a maximum of six official members. We recommend at least four (two programmers, one electrical and one mechanical)

3.4 In circumstances where the number of teams becomes unmanageable for the competition (around 15 teams per category), the organisational team will enforce that each university or representing group has a maximum of three teams allowed to compete.

3.5 Teams must be able to show their progress through a video diary/showcase/logbook before the competition to ensure they are ready to compete.

3.5.1 The logbook must be emailed to qutroboticsclub@gmail.com by 27/06/2025.

3.6 Mentors are allowed but must not have significant input in the team.

4. Category Restrictions

4.1 Teams within the Newcomers category must follow these criteria:

4.1.1 At least 75% of team members must not have previously competed in the DRC.

- (1) The exception to this is if a returning member has competed previously but achieved no successful track completions (i.e. no robot movement, complete system failure etc.), they still qualify as a 'newcomer' and do not count as a returning member.
- (2) Example team makeups following this rule are as follows: a 3-5 person team with up to 1 returning member, a 6 person team with up to 2 returning members.
- (3) Returning members are allowed as mentors but they must not have significant input on the team or droid (see *Section 3.6*)

4.1.2 Be a current undergraduate university or high school student.

4.2 Entry to the Opens category is not restricted. A first time team may choose to enter in the Opens category if they wish to. It is encouraged that first time teams enter in a category they feel reflects their level of expertise, for example, a first time team made up of final year engineering students may choose to enter in the Opens category rather than Newcomers due to their prior experience. It is strongly advised that all high school competitors enter into the Newcomers category given that they qualify.

4.3 A team may be moved from the Newcomers category to the Opens category in advance of competition day at the discretion of the organisational team in consultation with the team.

4.4 If there are not enough teams in either category, the categories will be merged so Newcomers and Opens teams will compete together at the discretion of the organisational team.

5. Robot Specifications

- 5.1 The total budget for the finished robot must not exceed \$1500 AUD. This is to ensure an even playing field. All components will be at the expense of the team.
- 5.2 The droid must be able to wirelessly start, stop and finish the race. Do this through using a handheld device that can control those functionalities of your robot. This device must not be used for any other parts of the race.
- 5.3 Any software may be used.
- 5.4 Any processing method, including off-board may be used.
 - 5.4.1 External devices used will not be included in the budget, provided they were not bought specifically for DRC.
- 5.5 Only on-board sensors may be used. Main guidance must be computer visionbased (i.e. No GPS, RC, LiDAR). The droid must use computer vision to sense the boundaries, obstacles and other droids.
- 5.6 The droid must be battery powered, i.e. no gas or fuel liquid powered propulsion.
- 5.7 Each robot must have a solid red outer shell that covers at least 80% of the robot.
- 5.8 Collision avoidance with other droids is not mandatory, however droids must not intentionally interfere with other droids in any way.
- 5.9 The droid must be a wheeled robot that remains on the ground the entire time.
- 5.10 LEDs or screens on the robot may be used as long as no other rules are broken.
- 5.11 Maximum Dimensions are as follows:
 - 5.11.1 Length: 800mm
 - 5.11.2 Width: 400mm
 - 5.11.3 Height: 350mm
- 5.12 It is recommended to avoid having an open wheel design. Two robots failing this may get their wheels tangled up. Please ensure that judges

can still see when a wheel has contacted the track edges or markers in your design.

- 5.13 Each robot will be judged on the day to ensure it meets specifications. If it does not, it will not be allowed to compete. At the beginning of the competition, robots will be scrutinised to confirm this. Any significant physical changes to the robot must be reported to the judges at the start of each heat to allow for further scrutineering.
- 5.14 At each heat, the team must make it clear to the judges that the robot is running autonomously, i.e. not using a remote control or keyboard to drive the robot directly. Any non-autonomous movement will result in a disqualification for that heat.

6. Track and Environment Specifications

- 6.1 The competition will be held at QUT Gardens Point Campus.
- 6.2 The track will be indoors on a hard flat surface.
- 6.3 The track will be in a large lab space meaning there may be equipment around that we cannot move (including objects the same colour as the track lines).
- 6.4 Yellow and blue coloured tape will indicate the track boundaries, with yellow tape on the left and blue tape on the right. (See *Appendix A*).
- 6.5 The start/finish line will be indicated by a line of green coloured tape. The direction of motion for the track may also be indicated by an arrow of the same green tape on one side of the start/finish line wherever is appropriate. See *Figure 1* for illustration. (See *Appendix A*).

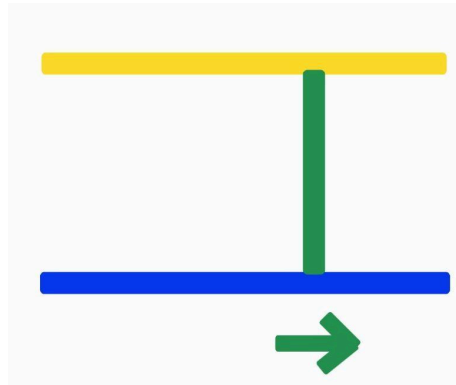


Figure 1. Green direction arrow next to start line

- 6.6 The same green tape (See *Appendix A*) may be used to mark the 25%, 50% and 75% track completion points on the track. These may be placed either on or next to the yellow or blue line, but may never occlude/break the line completely. See *Figure 2* for illustration.

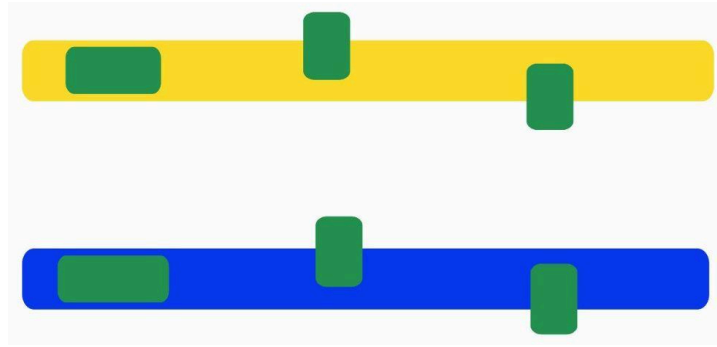


Figure 2. Green percentage completion markers on the track (for judging purposes)

- 6.7 The width of the track will range between one and two metres throughout.
- 6.8 In the **Opens category only**, there will be obstacles with the following specifications:
 - 6.8.1 They will vary in size (Maximum 400x400x500mm), and are painted purple. (See *Appendix B*)
 - 6.8.2 There may be up to 2 obstacles at any track section. The obstacle configuration will increase in difficulty with each round.
- 6.9 Each track will have a Turning Challenge (see *Section 8*) which includes a black arrow on the ground. This arrow presents a <5mm raised area on the track which has a different surface to the rest of the track.
- 6.10 Each track will sit within a 7m by 8.5m area. An example track can be found in *Appendix D*.
- 6.11 Judges will be walking around the edges, on and over the track during races to ensure proper scoring is conducted. Judges may be wearing clothing or shoes similar in colour to the track lines, obstacles or other items needing to be detected on the track. Please ensure your droids can cope with this.
- 6.12 Spectators are allowed within 250mm of the track edge on one side. Be wary of them wearing clothing or shoes similar in colour to track lines, obstacles or other items needing to be detected on the track.
- 6.13 Access to power and a table to set up will be provided.

- 6.14 Access to a workshop will be provided. It will have power supplies, soldering equipment, hot glue guns, basic hand tools, etc. The possibility of laser cutting and 3D printing is also available.
- 6.15 Teams should bring whatever non-workshop equipment they may need (Laptops, spare parts, etc.). We may be able to supply some items at request but cannot guarantee it.
- 6.16 Teams are encouraged to print spares of mechanical and electrical parts for the worst-case scenario.

7. Turning Challenge

- 7.1 A left- or right-hand turn road marking, similar to actual road markings, will be added to the track. This will indicate a left or right hand turn to be performed by the droid (these will be placed on the ground of the track). This will be placed within a specific shaped track element, as shown in *Figure 3*.

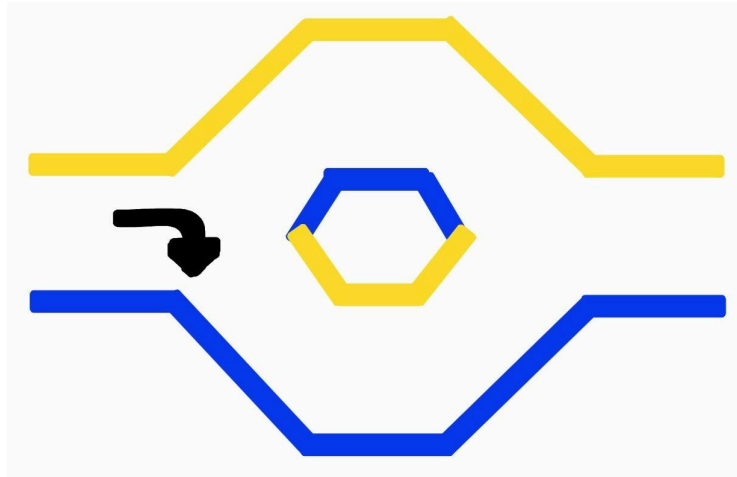


Figure 3. Turning challenge general layout

- 7.2 Only one turning sign shall be present on the track at any time.
- 7.3 The sign will be approximately 28x20cm and will be black.
- 7.4 The sign may be present up to a 3-4mm raised area on the track and its surface may also increase or decrease a droid's grip on the track. Droids will need to account for this.
- 7.5 The sign will indicate which side of the fork the droid shall take when it approaches the sign.
- 7.6 See *Section 8.4.3* for how the Turning Challenge is scored.
- 7.7 The direction of the arrow will be randomised with each race.
- 7.8 Both routes will be the same length and design
- 7.9 See *Appendix C* for arrow specifications.

8. Race Rules

- 8.1 Judges will scrutineer the droids before each race to ensure they comply with the rules.
- 8.2 When the heat is due to start, a judge will give a countdown for the heat to start. After which, both teams may hit the start button on their external device for their droid to start moving.
- 8.3 Droids will start nose on the start/finish line. Droids must stop once their nose reaches the start/finish line after a lap before being repositioned for another lap time to be recorded.
 - 8.3.1 Droids may be stopped manually via the team's external device.
 - 8.3.2 Droids which stop autonomously will receive a *time bonus of -5 seconds*.
- 8.4 During a heat, droids will have time penalties applied due to crossing track lines, hitting obstacles etc. as follows:
 - 8.4.1 Boundaries:
 - (1) One wheel goes outside boundaries of track or touches boundary tape -> *time penalty of +2 seconds*
 - (2) Multiple wheels go outside boundaries of track or touch boundary tape -> *time penalty of +5 seconds*
 - (3) Whole droid crosses a boundary/line of tape -> *time penalty of +10 seconds*
 - 8.4.2 Collisions:
 - (1) Droid has a collision with an obstacle, another droid or other object after which it continues driving -> *time penalty of +5 seconds*
 - (2) Droid has a collision with an obstacle, another droid or other object after which it stops -> *time penalty of +20 seconds*
 - 8.4.3 Turning Challenge:
 - (1) Droid takes the wrong turn at the Turning Challenge -> *time penalty of +5 seconds (Opens only)*

- (2) Droid takes correct turn at the Turning Challenge -> *time bonus of -5 seconds*
- (3) Droid is unable to choose a direction at the Turning Challenge -> *time penalty of +10 seconds (Opens only)*

8.5 **Depending on which category the team is competing in, certain time penalties may not be applicable**, as seen in the list above. To specify:

8.5.1 Newcomer teams are not penalised for inability to choose a direction at the Turning Challenge.

8.5.2 Newcomer teams will not have obstacles on their competition tracks, therefore time penalties 8.4.2.(1) and 8.4.2.(2) only apply to collisions with other objects such as environmental hazards and other droids.

8.6 Teams will have a certain amount of time to do as many laps as possible during each heat. (Time to be specified on the day based on the number of competing teams.)

8.7 Teams will have a certain amount of time to share the track between all teams for testing between heats. (Time to be specified on the day based on the number of competing teams)

8.8 The best time from the laps completed during each heat will be used to rank the teams, i.e. if there are four heats, each team will have their best time taken from each heat to figure out their final best time for team rankings and finals.

8.9 If a judge determines that a droid has entered an unrecoverable state (such as completely lost the track):

8.9.1 The car may be reset with time still running.

8.9.2 The team may choose to stop the heat, having the approximate percentage of the track completed and their current time recorded. Their score for that heat will be: $\text{Score} = (100\% - \% \text{ completion}) * \text{max heat time} + \text{current time}$

For example a team completing 20% of the track in 30 seconds in a 10 minute heat slot receives: $80\% * 600\text{s} + 30 = 510 \text{ seconds}$.

- 8.10 In the event that a droid is unable to compete in a heat, they will receive the maximum amount of time allowed for that heat as their score - eg. if the maximum time per heat is 10 minutes (600 seconds), they will receive a score of 600. This additionally applies if the team is disqualified for the round due to non-autonomous operation.

9. Design Judging

- 9.1 On competition day, each team will have a scheduled interview with a panel of expert robot design judges, within which they will present their droid for scoring.
- 9.2 It is recommended that each team have prepared a short (less than 5 minutes) presentation to present to the judges. This can include a powerpoint if desired, however this must be managed on your own device.
- 9.3 Design will be scored by each judge individually on a point scale in the following categories:
 - 9.3.1 Innovation - maximum 10 points
 - 9.3.2 Physical Robot Design - maximum 30 points
 - 9.3.3 Aesthetic - maximum 10 points
 - 9.3.4 Presentation Skills / Communication - maximum 10 points
- 9.4 The judge's scores go towards awarding the best robot design award for each category and contribute to the overall competition winner, as detailed in Section 11.
- 9.5 A detailed summary of judge feedback can be supplied after the competition on request, please email us at qutroboticsclub@gmail.com.

10. Finals Judging

10.1 The finals races to determine 1st Place, 2nd Place Runners Up and 3rd Place Runners up in each category will be conducted as follows:

10.1.1 At the end of the heats, all teams will be ranked in each category. The top 4 teams from each category will be entered into semi-finals.

10.1.2 In each semi-final, two teams from the top 4 will race around a double-length finals specific track.

10.1.3 The top 2 fastest teams in the semi-finals will then face off in a finals race with both team droids on the double-length finals specific track.

11. Prizes

Due to the uncertainty of changing the competition to a new 2-category system this year, we are still determining what prize money will be awarded for each of these prizes. Please bear with us as we figure this out.

- 11.1 A prize will be awarded to the overall winner based on a race between the 1st place Newcomers and 1st place Opens team. This race will not include obstacles in order to accommodate the Newcomers rules changes.
- 11.2 For the Opens category, placing prizes will be awarded as follows:
 - 11.2.1 1st Place
 - 11.2.2 2nd Place Runner Up
 - 11.2.3 3rd Place Runner Up
- 11.3 For the Newcomers category, placing prizes will be awarded as follows:
 - 11.3.1 1st Place
 - 11.3.2 2nd Place Runner Up
 - 11.3.3 3rd Place Runner Up
- 11.4 A prize will be awarded to the best designed droid overall, which will be judged by an expert panel. See *Section 9*.
- 11.5 A prize will be awarded for the organisers choice in both categories combined. This will be determined by the QUT Robotics Club Executive.

Appendices

Appendix A: Tape

Official Blue Tape:

https://www.bunnings.com.au/scotchblue-36mm-x-55m-original-multi-surface-painter-s-masking-tape_p1214159

Official Yellow Tape:

<https://www.southeastclearance.com/products/yellow-painters-masking-tape?variant=44137120825587>

Official Green Tape:

https://www.bunnings.com.au/bear-24mm-x-50m-green-painters-edge-masking-tape-24mm_p0144255

Appendix B: Obstacles

Official Colour of Obstacles: Magic Magenta by Taubman's

<https://encycolorpedia.com/864e79>

Example Obstacle:

https://drive.google.com/drive/folders/1h_r-iln--jWyq43rSTpdbEBnNRCACWT-?usp=share_link

Appendix C: Arrow

Official Information for the turning arrow:

https://drive.google.com/drive/folders/1CeoVitrw1RynVSLZrAoV2ZQFHG5e3hgck?usp=share_link

Appendix D: Example Track

Example track from 2024 competition:

